

Skills Summary

Solving Exponential Equations with Common Bases

Use this reference while completing your worksheet or when studying.

Skill 1 — Equations that already share a base

One-to-one property. If $b > 0$ and $b \neq 1$, then $b^M = b^N$ exactly when $M = N$.

So once both sides are powers of the same base, the exponents become an ordinary equation.

Example: Solve $2^{3x-1} = 2^{x+7}$.

Step 1. Both sides are already powers of 2, so the one-to-one property turns this into a linear equation in the exponents.

Step 2. $3x - 1 = x + 7$, so $2x = 8$.

$$\Rightarrow x = 4$$

Try it: Solve $5^{4x} = 5^{x+9}$.

check: $x = 4$

Skill 2 — Rewriting both sides to one common base

Tools for matching bases: $(b^m)^n = b^{mn}$, $\frac{1}{b^n} = b^{-n}$, $\sqrt[n]{b} = b^{1/n}$.

Pick the *smallest* base that generates both sides (4, 8, 16, 32 are all powers of 2; 9, 27, 81 are all powers of 3).

Example: Solve $4^{x+1} = 32$.

Step 1. Neither side is a power of the other, but 4 and 32 are both powers of 2 — rewrite with base 2 first.

Step 2. $(2^2)^{x+1} = 2^5$, so $2x + 2 = 5$.

$$\Rightarrow x = \frac{3}{2}$$

Try it: Solve $9^x = 243$.

check: $x = \frac{5}{2}$

(!) Watch out: $\left(\frac{1}{2}\right)^x$ is 2^{-x} , not 2^x . A reciprocal base flips the sign of the whole exponent, brackets and all.

Skill 3 — Reading an exponential model from a table

Constant ratio $\Rightarrow$ exponential. If each step in x multiplies $f(x)$ by the same number r , then $f(x) = f(0) \cdot r^x$.

To solve $f(0) \cdot r^x = V$, divide by $f(0)$ first, then match bases.

Example: A table gives $f(0) = 7$, $f(1) = 14$, $f(2) = 28$, $f(3) = 56$. Solve $f(x) = 448$.

Step 1. Every step doubles the value, so the ratio is 2 and the model is $f(x) = 7 \cdot 2^x$ — divide by the starting value to isolate the power.

Step 2. $2^x = \frac{448}{7} = 64 = 2^6$.
 $\Rightarrow x = 6$

Try it: $g(0) = 5$, $g(1) = 15$, $g(2) = 45$, $g(3) = 135$. Solve $g(x) = 405$.

check: $5 = 5$